



An optimized ensemble model based on meta-heuristic algorithms for effective detection and classification of breast tumors

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Abstract

One of the most common cancers among women worldwide is breast cancer (BC), and early diagnosis can save lives. Early detection of BC increases the likelihood of a successful outcome by enabling treatment to start sooner. Even in areas without access to a specialist physician, machine learning (ML) aids in early BC detection. The medical imaging community is becoming more interested in using ML, and deep learning (DL) to increase the accuracy of cancer screening. Many disease-related data are sparse. However, for DL models to perform well, a large amount of data is required. Because of this, the DL models that are currently in use on medical images are not as effective as they could be. Convolutional neural network (CNN) models have recently gained popularity in the medical industry, and they perform admirably in terms of high performance and robustness at image classification. The proposed method classifies data using ensemble pre-trained models such as the dense convolutional network (DenseNet)-121 and EfficientNet-B5 feature extractor networks, as well as the support vector machine for classification. Using a modified meta-heuristic optimizer, the selected pre-trained CNN hyperparameters were optimized to improve the performance. The experimental results for the presented model on the INbreast dataset show that the EfficientNet-B5 model is effective for BC classification, with overall accuracy, sensitivity, specificity, precision, and area under the ROC curve (AUC) values of 99.9%, 99.9%, 99.8%, 99.1%, 1.0, respectively.

Keywords Deep learning · Optimization · Transfer learning · Breast cancer · Convolutional neural network

1 Introduction

According to a recent “American Cancer Society” [1] report, 31% of newly diagnosed instances of cancer in women are in BC, as illustrated in Fig. 1. It is the most common cancer in women worldwide, with an extremely high death rate. Currently, the most efficient way to decrease BC deaths is through early detection and treatment. Immunohistochemistry (IHC) is a diagnostic analysis technique commonly used by pathologists to ascertain the hormone receptor status and histopathological grade. As digital pathology images, which are frequently full slide images (WSIs), accumulate, the cancer diagnostic becomes extremely time-consuming and vulnerable to variations in observation [2, 3].

DL, a subset of artificial intelligence (AI), applies artificial neural networks (ANNs) to stimulate the human brain and learn from enormous data amounts. DL models are favored over conventional ML algorithms as they can

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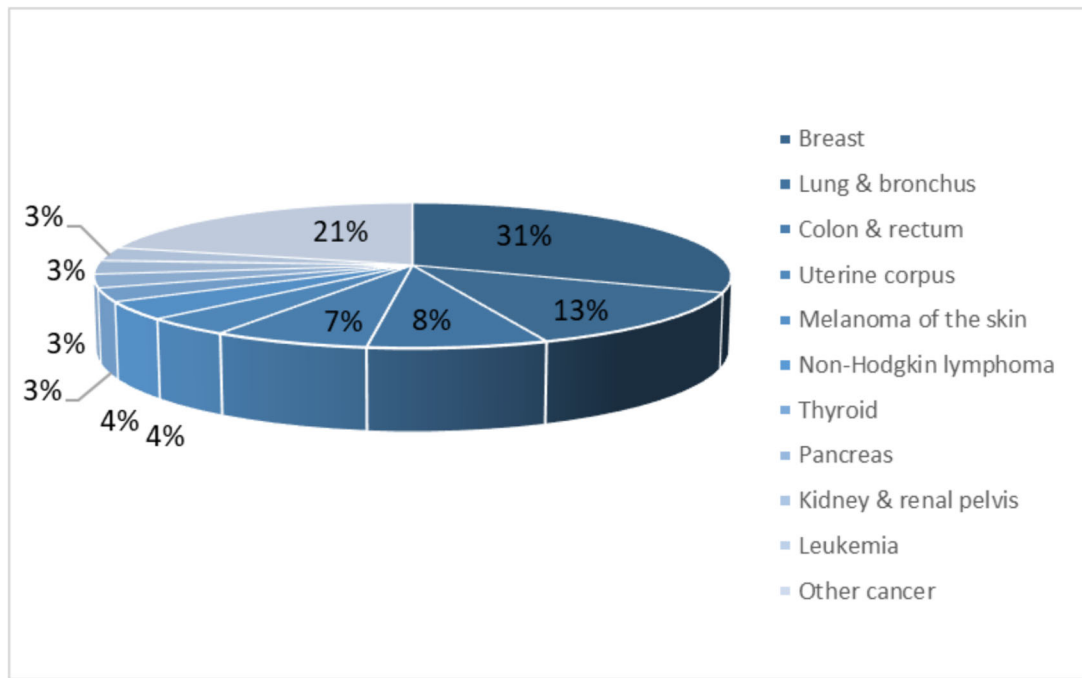


Fig. 1 The estimation of new cancer cases in 2023

concentrate on intricate image features and learn independently [4]. Many medical fields have used DL techniques, and new models are always being created to enhance the extraction of features. Some of the advantages of DL include handling large amounts of data, improving time efficiency, mastering unstructured data, advanced analytics, and cost-effectiveness. Diverse methods are commonly employed for tasks such as image processing, segmentation, and classification, referred to as DL techniques [5, 13].

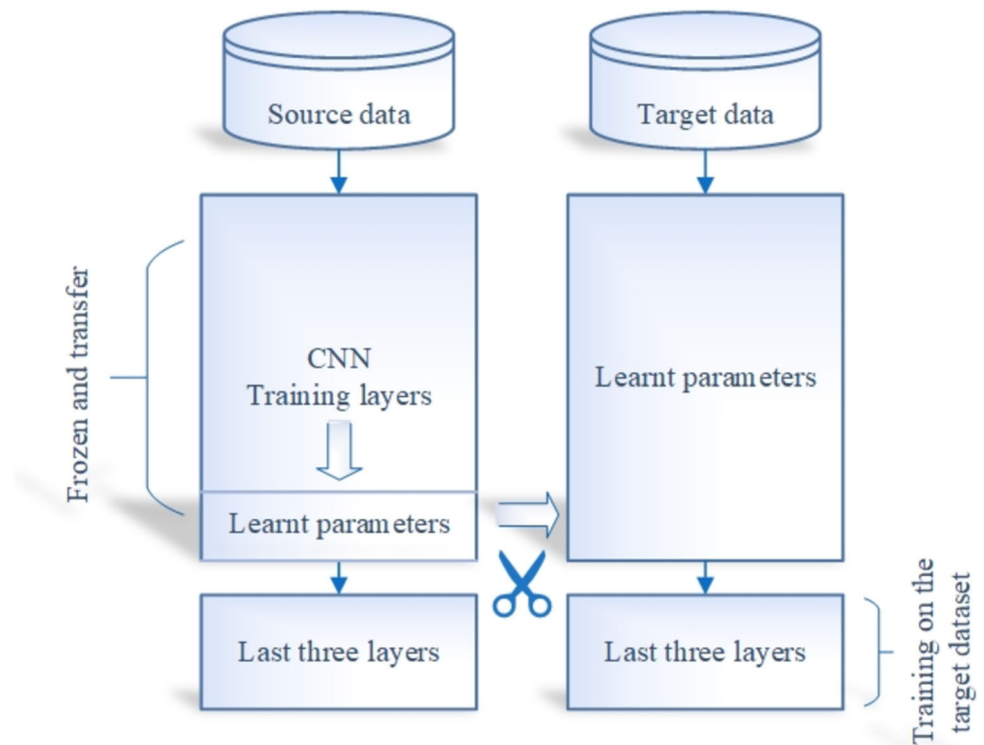
Several imaging applications have seen the successful application of DL approaches in recent years, according to a review of recent papers on DL applications. In medical image analysis, CNN models have demonstrated efficacy in cell recognition, tumor detection, and skin disease classification [5, 6]. These designs use layer-based feature extraction, and the values of a CNN model's hyperparameters determine how well the model performs. Finding these hyperparameters' ideal values can be challenging, time-consuming, and need experimentation. Many medical images pose an NP-hard optimization challenge when it comes to tuning hyperparameters [7, 8]. A CNN training process requires a large data amount, yet medical data is scarce. This problem can be resolved by using transfer learning (TL) from a natural-images dataset such as ImageNet along with a fine-tuning approach, as Fig. 2 shows. Performance can be enhanced by applying the TL concept, which combines the expertise of many CNN

designs. The principal advantages of TL are enhanced classification accuracy and expedited training.

When utilized to address classification issues, heuristics are very appealing and have been applied to numerous real-world issues [9, 10]. Meta-heuristic algorithms are effective tools for optimizing the parameters and architecture of DL models. It can precisely calculate the appropriate estimation of DL components, such as hyperparameters, weights, number of layers, and learning rate.

The Grey Wolf Optimizer (GWO) algorithm has been tested on 29 well-known test functions and performed comparably to other meta-heuristics, such as particle swarm optimization (PSO), gravitational search algorithm (GSA), and differential evolution. It has also effectively tackled traditional engineering design challenges and real-world optimization problems in numerous disciplines. Also, the Hunger Games Search (HGS) method outperforms other well-known algorithms on 23 functions and meets IEEE CEC 2014 benchmarks. It has been used to solve various engineering difficulties, demonstrating its adaptability. The HGS algorithm excels at solving continuous and discrete optimization problems, making it a valuable tool. Multimodal optimization issues, in which the search space contains many local optima, can also be handled by the HGS method.

The presented study proposes a new model for multi-class BC detection and classification. The current model consists of three major components (data collection and preprocessing, choosing the best hyperparameter values,

Fig. 2 Transfer learning technique

and model training and evaluation) used to improve breast pictures and thus classification outcomes. These stages are explained in depth in the subsections that follow.

The main motivating factors for this work are outlined below.

- 1) A novel hyperparameter tuning technique that uses TL and an improved meta-heuristic algorithm is presented.
- 2) We conduct a comprehensive comparison of optimizing the hyperparameters of the proposed CNN using the HGS and GWO algorithms.
- 3) Remove non-breast regions in preprocessing steps to reduce the training computation time.
- 4) Detecting and classifying breast mammography more precisely using the SVM classifier.
- 5) Our results provide an exciting opportunity for more research and useful applications, which could lead to more quick and accurate diagnostics and ultimately enhance the quality of life for those suffering from BC.

The paper has been laid out as follows: Sect. 2 discusses the associated work, and Sect. 3 presents the proposed model. The experimental results are compared with the real data in Sect. 4. Section 5 brings the context conclusion.

2 Related works

These days, AI knows no bounds and is transforming and saving lives. AI works to improve and streamline interactions between patients and providers by simulating human intelligence through computer systems and consists of the capacity to investigate and solve issues. AI and technology are advancing applications in healthcare, such as decision support, image analysis, and patient triage. Computer-aided diagnosis (CAD) acts as a second set of eyes, indicating anomalies (such as a tumor or micro-calcification) on digital mammograms, and advises the radiologists reading the images to take appropriate action or not. Radiologists with CAD performed as though two radiologists were double reading. Experts and company owners have created CAD systems to automatically classify BC as benign or malignant. Choosing the best algorithms for a CAD system necessitates improving comprehension of the information included in cancer images [11].

Abeer et al. [12] developed a deep CNN model, which extracted features from the MIAS dataset using pre-trained CNNs. It performs well in the six-evaluation metrics. The VGG16 model outperforms other models in BC diagnosis, with overall accuracy and precision of 98.96% and 98.87%, respectively.

Using the INbreast dataset, Akselrod-Ballin et al. [13] assessed a DL model for BC classification using a region-based CNN. The suggested model had an accuracy rate of

78%. On the same dataset, Al-Antari et al. [14] constructed a DL technique to classify BC using feedforward CNN, ResNet 50, and Inception ResNet-V2 CNNs. The accuracy of the suggested model was 95.32%.

Chakravarthy et al. created a TL-based framework [15]. The last 3 layers, in AlexNet, GoogleNet, ResNet-50, and Dense-Net 121 networks are trained using the INbreast dataset. The SVM classifier is applied to categorize the breast tumor (BT) classes. The proposed model obtained a 96.6% accuracy rate.

Abeer et al. [16] developed a DL technique that depends on pre-trained CNNs for detecting and classifying BC using the INbreast dataset. Breast mammographic images are preprocessed to enhance the images and minimize computation time. The model achieved accuracy, sensitivity, specificity, and AUC values of 97.1%, 96.3%, 97.9%, and 0.988%, respectively.

Abeer et al. [17] developed a DL architecture combining TL and long short-term memory for automatic BC detection and diagnosis using the 80–20 method. The model extracts the learned features from pre-trained CNNs, and extracts features from the INbreast dataset. The presented model achieved high accuracy, sensitivity, precision, and AUC values with 99.236%, 99.1%, 96%, 98.8%, and 0.998, demonstrating its effectiveness in detecting breast tumors.

Abeer et al. [18] developed a new DL to detect and classify tumors in mammographic images using TL techniques. The model uses pre-trained networks from VGG-19, VGG16, and InceptionV3, to decrease training time and raise classification accuracy. The Mammographic Image Analysis Society dataset is applied for feature extraction. The model outperforms VGG19 and Inception V3 in accuracy, sensitivity, and specificity.

Sahu et al. [19] illustrated a DL-based ensemble classifier is proposed for BC detection, demonstrating efficient performance with a small dataset. This method integrates TL models and employs residual learning, depthwise separable convolution, and inverted residual bottleneck structure. Experimental results show superior classification performance, with 99.17% accuracy for abnormality and 97.75% for malignancy on the mini-DDSM dataset and 94.62% accuracy on BUSI and BUS2, respectively.

Qian et al. [20] presented a novel model to identify atypical BC using the ZFNet CNN for breast mammographic images. The method uses Wiener and CALHE filters for preprocessing, modified and trained on the CBIS-DDSM dataset, and an extreme learning machine to enhance classification performance. The method uses the Chimp Optimization Algorithm (SWChOA) and benchmark meta-heuristic optimization algorithms. When compared to other approaches, the performance level of the provided methodology was the greatest.

Gudur et al. [21] developed a hybrid genetic algorithm and CNNs to optimize CNN architecture for BC categorization. This dynamic optimization process extracts relevant features and patterns from histopathology pictures, improving diagnostic precision. Tests show significant improvement in diagnosis accuracy and reduced manual hyperparameter adjustment. This combination of genetic algorithms and CNNs enhances BC diagnosis and treatment outcomes with 98.54%, 99.41%, 98.42%, and 97.02% for accuracy, precision, recall, and F-Score, respectively.

Rahman et al. [22] presented a computational model for BC diagnosis using a ResNet-50 CNN. The framework classifies mammogram images using TL, achieving an impressive 93% classification accuracy. This approach could save lives and resources by reducing error rates in screening mammograms.

Srikantamurthy et al. [23] created a hybrid CNN-LSTM model to categorize four BC subtypes on a dataset as benign and malignant, reaching 92.5% and 99% overall accuracy in multiclass and binary classification, respectively.

Sharmin et al. [24] developed a hybrid BC detection approach that integrates DL with ensemble-based ML methods. Experiments on a publicly available dataset show the model's robustness and high performance compared to state-of-the-art models, achieving accuracy, precision, recall, and F1 scores of 95%, 94.86%, 94.32%, and 94.57%, respectively.

3 Proposed model

This section presents the proposed novel ensemble model based on meta-heuristic algorithms GWO and HGS algorithms for hyperparameter optimization-based BT classification. To achieve the best accuracy, the meta-heuristic algorithms select the optimal hyperparameter values of the networks. The structure of a high-level CAD system for cancer diagnostics is shown in Fig. 3. Breast tumors are classified as benign or malignant using four major processing stages. The presented model contains three phases as shown in Fig. 4, and operates in the following order:

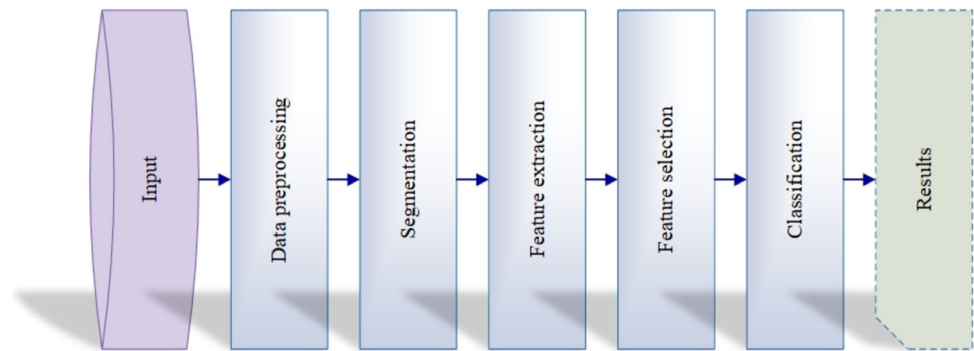
1. Data collection and preprocessing
2. Choose the best hyperparameter values
3. Model training and evaluation

3.1 Data collection and preprocessing

1. Noise removal.

The best techniques for eliminating noise from mammography pictures are the wavelet transform and the

Fig. 3 The general architecture for CAD systems



median filter [25]. The 3×3 median filter is used in this study to remove noise from the mammographic images.

2. Morphological analysis.

Morphological analysis is a collection of image processing methods that manipulate images depending on their forms. By adding a structural element (SE) to an input image, morphological operations yield an identically sized output image. The result of a morphological operation determines each pixel's value by comparing it to its neighbors in the corresponding pixel in the input image. Morphological procedures, like spatial filtering, entail moving the structuring element across every pixel in the source image to produce a pixel in the altered new image. The morphological operation that is carried out determines the value of this new pixel [26]. The operations of morphological analysis are illustrated in Fig. 5.

3 Constructed limited histogram equalization.

One type of adaptive histogram equalization that prevents over-amplification of noise is called Constructed Limited Histogram Equalization (CLAHE). Using CLAHE, an image is divided into small areas known as tiles, and each tile is subjected to a unique application of histogram equalization. The histogram is then clipped at a predefined threshold value to limit the contrast in each tile. The clipped portion of the histogram is then uniformly redistributed over all histogram bins. An image with enhanced edge definition and increased local contrast in every area of the image is the result of this procedure [27].

4 Augmentation.

Rather than gathering more data, image augmentation can be used to improve model classification accuracy [28]. Thus, the effectiveness of the proposed model is improved by the data augmentation approaches that increase the number of samples. Several data augmentation methods were used on the training dataset to raise the model's accuracy. These covers flipping both horizontally and

vertically, shearing, width and height shifts, and brightness and rotation modifications as presented in Algorithm 1.

Algorithm 1 Breast data augmentation (BDA).

Input: - Training dataset (ψ) Procedure: For each sample η in ψ do $\delta \leftarrow \text{Flip horizontally}(\eta)$ $\delta \leftarrow \text{Add brightness}(\eta, (0.5, 1.5))$ $\delta \leftarrow \text{Shear}(\eta, 0.2)$ $\delta \leftarrow \text{Rotate}(\eta, 60)$ Output: - Augmented data (δ)
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3.2 Choose the best hyperparameter values

A CNN training process needs a large data amount, and medical data is scarce, particularly in breast tumors. This issue can be fixed by combining TL from a natural-images dataset like ImageNet with a fine-tuning strategy. Applying the TL approach, which integrates the knowledge of several CNN designs, can improve performance. Improved classification accuracy and quicker training are the two main benefits of TL. A model transfer, whereby the target domain's network characteristics are applied, performance-optimized, and pre-trained using the source data, is a suitable technique for training models.

A. Grey wolf optimization algorithm.

A population-based meta-heuristic algorithm known as the Grey Wolf Optimizer (GWO) simulates the actual hunting and leading patterns of grey wolves. The program simulates the leadership hierarchy using 4 sorts of grey wolves: alpha, beta, delta, and omega. To optimize, the

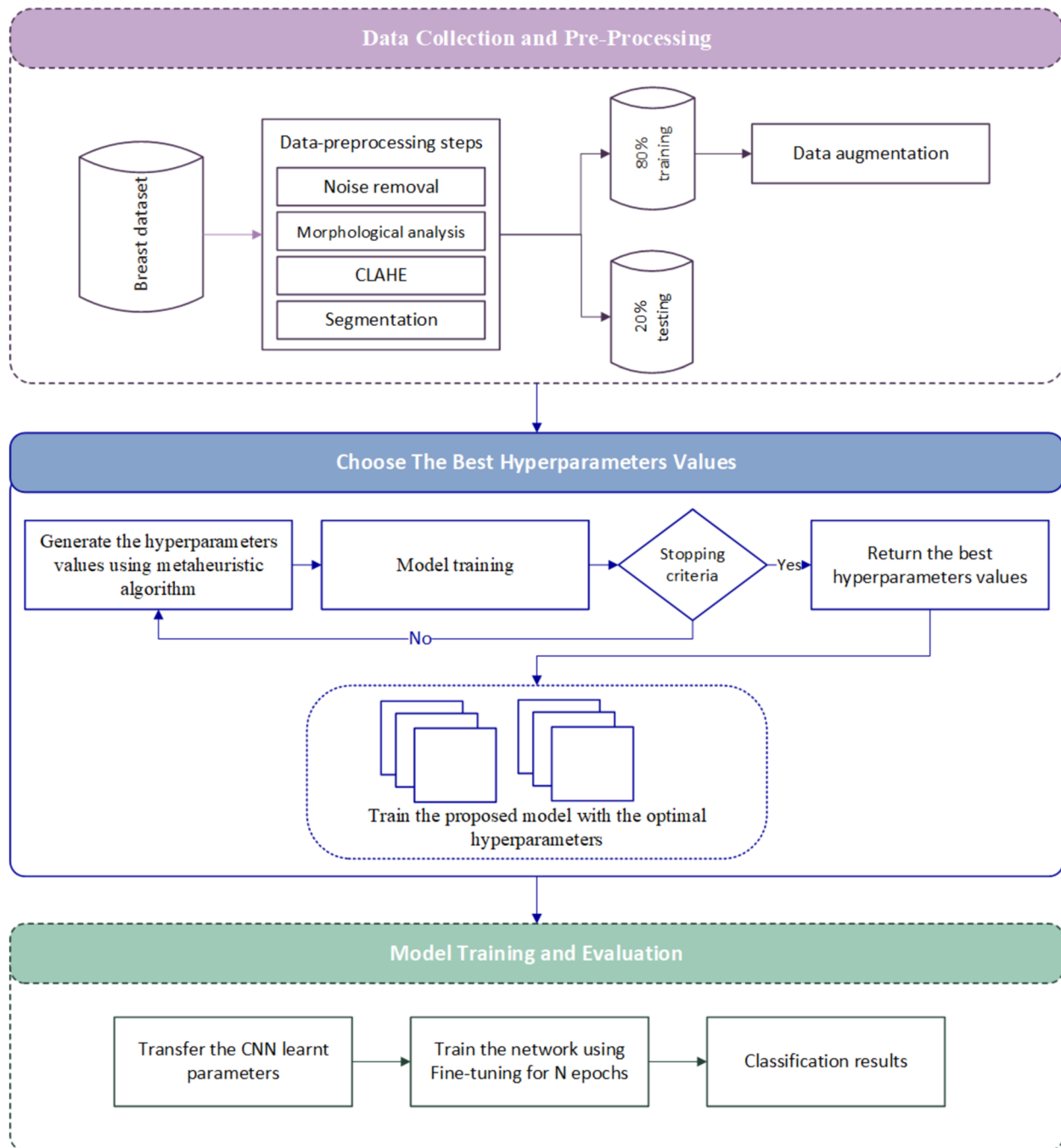


Fig. 4 The proposed model

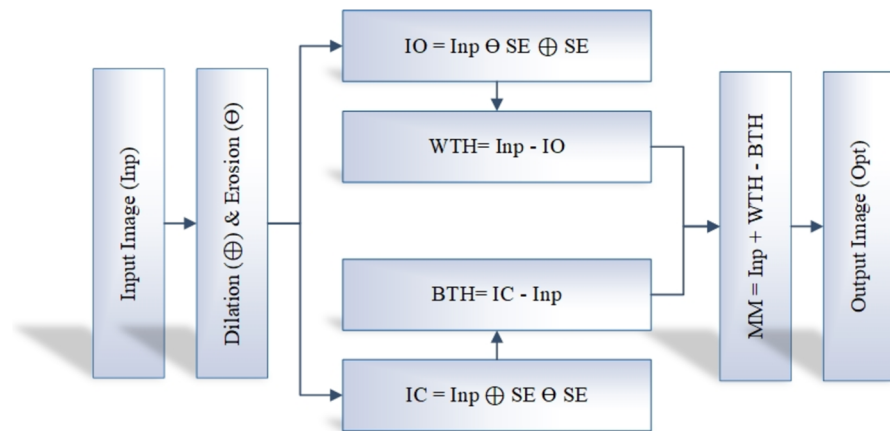
algorithm comprises three key hunting steps: searching for prey, encircling prey, and attacking prey.

The GWO is a nature-inspired meta-heuristic algorithm that replicates the natural leadership hierarchy and hunting mechanism of grey wolves. To replicate the leadership hierarchy, the algorithm employs four kinds of grey wolves: alpha, beta, delta, and omega. To perform optimization, the algorithm has three main hunting steps: searching for prey, encircling prey, and attacking prey. The

GWO algorithm starts with a population of wolves, each representing a possible solution to the optimization issue. The algorithm then iteratively modifies each wolf's position based on some performance metric until it reaches an optimal solution.

- Social hierarchy

The GWO method is based on a mathematical model of the wolf social order. The strongest solution is alpha (α),

Fig. 5 The morphological analysis operations

Where (IO = Image Opening, IC = Image Closing, WTH= White Top-hat, BTH = Black Top-hat)

followed by beta (β) and delta (δ), which are the second and third-best solutions, respectively. The remaining solutions are referred to as omega (ω). The alpha, beta, and delta wolves lead the optimization process, while the omega wolves follow. This is the foundation of the GWO algorithm's optimization approach.

- Encircling prey

The following equations can be used to mathematically model grey wolf encircling behavior during hunting:

$$\vec{D} = |\vec{C} \cdot X_p(t) - \vec{X}(t)| \quad (1)$$

$$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \cdot \vec{D} \quad (2)$$

where t refers the current number of iterations, $\vec{X}_p$ denotes the prey location vector, $\vec{X}$ denotes the location of a single wolf, and A and C can be calculated by the following:

$$\vec{A} = 2\vec{a} \cdot \vec{r}_1 - \vec{a} \quad (3)$$

$$\vec{C} = 2 \cdot \vec{r}_2 \quad (4)$$

As the number of iterations increases, the vector is reduced linearly from 2 to 0. $\vec{r}_1$ and $\vec{r}_2$ are random vectors with values ranging from 0 to 1.

- Hunting

Grey wolves have the natural ability to locate their prey and circle them to get a better look. However, it is impossible to pinpoint the precise location of the optimum (prey) in an abstract search space. To mathematically model the hunting behavior of grey wolves, the GWO algorithm considers the alpha as the best solution, with beta and delta also having some knowledge about the potential location of prey. As a result, the top 3 solutions are saved, and the remaining search agents (omegas) must alter their locations to reflect these positions. To model this behavior,

the following formulas are suggested. The equations below are applied to calculate the positions of the leaders.

$$\vec{D}_\alpha = |C_1 \cdot \vec{X}_\alpha - \vec{A}| \quad (5)$$

$$\vec{D}_\beta = |C_2 \cdot \vec{X}_\beta - \vec{A}| \quad (6)$$

$$\vec{D}_\delta = |C_3 \cdot \vec{X}_\delta - \vec{A}| \quad (7)$$

$$\vec{X}_1 = |\vec{X}_\alpha - \vec{A}_1 \cdot \vec{D}_\alpha| \quad (8)$$

$$\vec{X}_2 = |\vec{X}_\beta - \vec{A}_2 \cdot \vec{D}_\beta| \quad (9)$$

$$\vec{X}_3 = |\vec{X}_\delta - \vec{A}_3 \cdot \vec{D}_\delta| \quad (10)$$

$$\vec{X}(t+1) = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3} \quad (11)$$

$$\vec{X}(t+1) = w_1\vec{X}_1 + w_2\vec{X}_2 + w_3\vec{X}_3, \text{ where } \sum_{i=1}^3 w_i = 1 \quad (12)$$

- Exploration

Wolves have the instinct to identify their prey's location and move around it to encircle it. However, in an abstract search space, knowing the precise location of the optimum (prey) is impossible. The GWO considers alpha to be the best solution for mathematically modeling grey wolf hunting behavior, with beta and delta also having some knowledge about potential prey location. As a result, the first 3 best answers are kept, and the remaining search agents (omegas) must change their locations based on those of the best search agents.

- Exploitation

The grey wolves in the GWO algorithm mostly follow the alpha, beta, and delta but also deviate from them to

explore the search space. The coefficient vector $\vec{A}$ is used with random values greater than one or less than -1 to model this divergence behavior, which encourages search agents to explore globally. Another component that promotes exploration and avoids local optima is the use of $\vec{C}$ with random values in $[0, 2]$. This vector assigns random weights to prey, which can either emphasize or deemphasize their role in determining distance. In contrast to $\vec{A}$, the values of $\vec{C}$ are not decreased linearly, and random values are used throughout the optimization process to emphasize exploration in both the initial and final iterations. This component is particularly useful when the optimization process becomes stuck at local optima.

The detailed algorithm of the modified GWO- ensemble model is shown in Algorithm 2.

Algorithm 2 Pseudo code of the modified GWO- ensemble model. B.

Input: P : Population size, and T : Max number of iterations
Output: classify breast tumors.
dataset split: Partition the dataset (80% for training and 20% for testing)
GWO initialization: Produce a random population X_i , a , A , and C
For every individual **do**
 Address the ensemble model to correspond to the hyperparameters formulated as solution.
 Evaluate the fitness of X_i .
End for
Assign the 1st, 2nd, and 3rd global best solutions to $X\alpha$, $X\beta$, and $X\delta$.
Do while ($t < T$)
 for each individual in X_i **do**
 determine the new location of X_i
 Set the hyperparameters for the ensemble model to correlate with the solution.
 Determine the fitness of the solution, X_i , in relation to the MSE error.
 End for
 Update the parameters i.e., a , A , and C
 Update $X\alpha$, $X\beta$, and $X\delta$
 $t = t + 1$
End while
Set the ensemble model architecture depends on the hyperparameters founded from the solution set.
Perform the test set classification.
End

Hunger game search optimization algorithm.

The HGS algorithm is an optimization method based on meta-heuristics and motivated by foraging and hunger instincts of animals. It is designed to tackle optimization problems based on social behaviors, specifically emulating how hunger affects a single animal's behavior. The idea of hunger is included by the algorithm in the search process

by using adaptive weights for stimulating the hunger effect. A step-by-step explanation of how the HGS algorithm is presented as follows:

- **Initialization:** The algorithm generates random candidate solutions from the search space.
- **Hunger adaptation:** An adaptive weight generated from the concept of hunger is devised and applied to model the effect of hunger on the candidate solutions. This weight is updated dynamically throughout the search process, affecting the exploration and exploitation of the search space.
- **Fitness evaluation:** The fitness of each candidate solution is evaluated based on the objective function of the optimization problem. The fitness value indicates the quality of the solution.
- **Selection:** The selection process is based on the fitness values of the candidate solutions. The algorithm uses a fitness-proportional selection mechanism to choose the

best solutions for the next generation.

- **Crossover and mutation:** The selected candidate solutions undergo crossover and mutation operations to generate new candidate solutions for the next generation. These operations help to explore and exploit the search space more effectively.
- **Hunger update:** The adaptive weight is updated based on the hunger level of the candidate solutions. The

hunger level is defined by the distance among the present solution and the best solution found thus far. The higher the distance, the hungrier the solution, and the higher the adaptive weight.

- **Termination:** When a stopping criterion is reached, like arriving to the maximum iterations number or obtaining a good solution, the algorithm terminates.

The hunger game search optimizer for the suggested model is shown in Fig. 6.

The HGS algorithm has several advantages, including simplicity, high performance, adaptability, and efficiency in finding and developing the target solution space. The algorithm is based on the concept of hunger, which is a critical motivator for animals, and applies an adaptive

weight depending on the idea of hunger to influence each search step. This means that the search process is simulated in a way that mimics how animals search for food. The algorithm has been validated on a comprehensive group of benchmark problems and has demonstrated excellent intensification and diversification capabilities. The approach has been used to solve numerous optimization issues, including global optimization, feature selection, and engineering design, and has shown reliable performance and excellent accuracy. The algorithm is straightforward to understand and has many potential applications in various fields, including computer science, engineering, finance, and biomedicine.

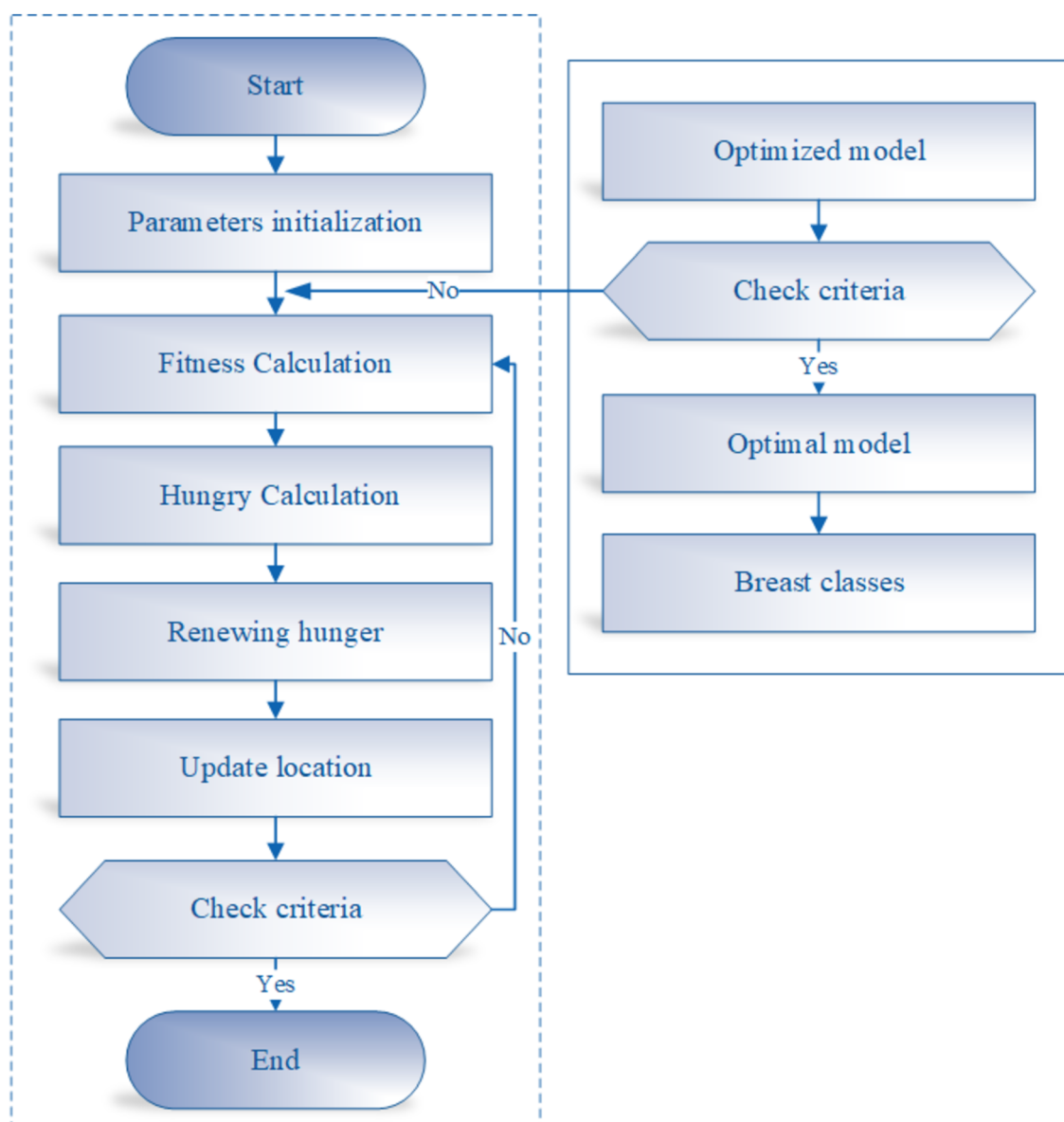


Fig. 6 The Hunger Game Search optimizer for the ensemble model

3.3 Model training and evaluation

In this study, feature extraction is carried out using the DenseNet-121, and EfficientNet-B5 networks. The ImageNet dataset is used to train this network. Colors, lines, and other input features are recognized by the filters in the network layers. Small parts and insignificant shapes can then be identified. The category that the input image is part of can be identified from the generated output. The pre-trained models are then used to categorize various objects in the new dataset. Excepting the final three layers, all trained parameters are frozen and carried over to the target task from the source task. The previously processed photographs are then utilized to continue network training.

As a result, there are very few newly trained dense layers. Furthermore, the previously trained layers in the pre-trained network are integrated with these layers to create a new class categorization. As a result, the training process can be completed fast, with minimal training data required when compared to CNN training from scratch. The collected features are then utilized to train an SVM classifier to perform classification tasks. The HGS algorithm is used to fine-tune the hyperparameters to find the best ones.

A CNN architecture, DenseNet-121 belongs to the DenseNet model family. A 121-layer deep neural network called DenseNet-121 is intended for use in image classification applications. It has been pre-trained on the ImageNet dataset, which contains over a million images across 1000 classes. It has several important qualities, like:

Dense blocks: Four dense blocks, each containing several convolutional layers, make up DenseNet-121. Each layer in a dense block is feed-forward connected to every other layer, enabling feature reuse, and mitigating the vanishing-gradient issue.

Transition layers: A transition layer decreases the amount of feature maps and their spatial dimensions among dense blocks.

Global average pooling: A global average pooling layer at the network's end averages the feature maps across the spatial dimensions to generate a single feature vector for every picture. After that, a fully connected layer receives this feature vector for classification.

Specifically, EfficientNet-B5 is trained at 456×456 resolution on ImageNet-1k. The novel scaling technique that underpins the model architecture uniformly scales the network's dimensions to improve performance. It provides the features and information listed below:

Model architecture: The EfficientNet-B5 model is built upon the EfficientNet family of models, which aim to provide higher performance and accuracy efficiency.

- **Training:** At a resolution of 456×456 , ImageNet-1k is used to train the model.
- **Pretrained weights:** For TL and fine-tuning tasks, EfficientNet-B5 has pretrained weights available.
- **Model usage:** Image classification tasks can be performed with EfficientNet-B5.

EfficientNet-B5 is renowned for its cutting-edge ImageNet accuracy as well as its efficiency in both inference speed and model size. For a variety of computer vision tasks, such as image segmentation, and image classification, it is an effective tool.

The efficacy and efficiency of the EfficientNet family of models, which includes EfficientNet-B5 in a range of computer vision applications, have led to a rise in popularity among DL experts.

SVM, a supervised ML technique, is utilized for regression and classification applications. SVMs have a wide range of applications, including handwriting recognition, spam detection, face detection, and anomaly detection. Because SVMs can handle high-dimensional data and nonlinear relationships, they are useful in a variety of applications.

4 Experimental results

1 Dataset.

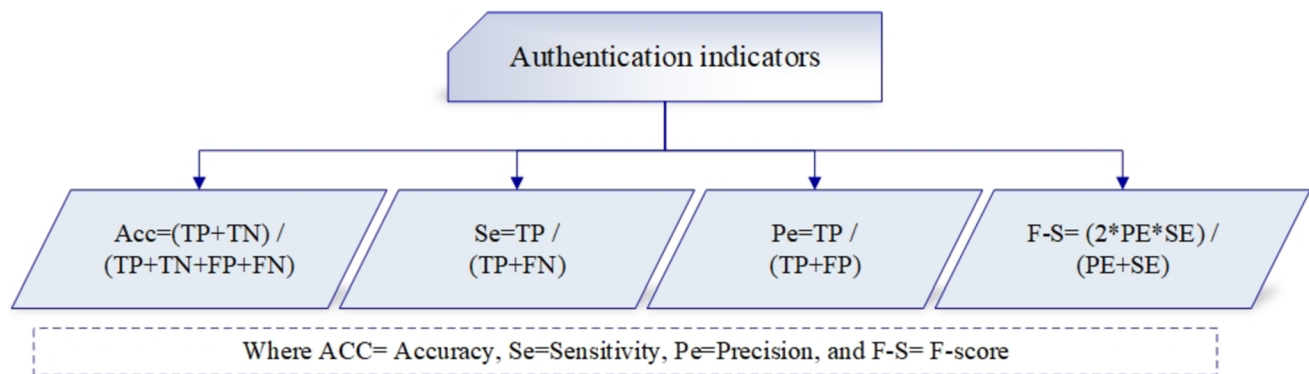
The INbreast dataset is a mammographic dataset with 115 cases (410 images), of which 25 cases (two images per case) are from women who only have one affected breast and 90 cases (four images per case) are from women who have both breasts affected. There are four different types of lesions in the dataset. The Key features of the INbreast are illustrated in Table 1.

2 Experimental results.

This section describes numerous tests to examine the proposed model's performance on the INbreast dataset. TL is used in an ensemble of DenseNet-121 and EfficientNet-B5 networks to improve classification results. The efficiency of the proposed ensemble model was tested using authentication indicators produced by the evaluation metrics for 3 classes, as illustrated in Fig. 7. The dataset was separated into three categories: benign, malignant, and normal. Then it was divided into 80% training and 20% testing activities. The advantages of preprocessing were evaluated by running experiments twice, before and after preprocessing. Tables 2 and 3 show the classifier's performance results before and after preprocessing for the proposed ensemble model. It has been noticed that data preprocessing can improve the classification results of breast mammography images.

Table 1 The INbreast dataset features

Key features of the INbreast dataset	
Total cases	It contains 115 cases (410) images
Lesions type	Masses, asymmetries, calcifications, and distortions
Contours	Created by experts and provided in XML file
Public availability	It is publicly available for research purposes at: https://www.kaggle.com/datasets/martholi/inbreast

**Fig. 7** The authentication indicators**Table 2** The INbreast dataset results were applied using the ensemble pre-trained CNN and GWO optimizer

CNN	Class	Performance of the classifier				
		Accuracy (%)	Sensitivity	Specificity	Precision	AUC
Before preprocessing	Benign	65.9	45.2	69.37	56.9	0.51
	Malignant	66.5	43.4	69.9	60.3	0.53
	Normal	68.1	49.8	68.4	60.9	0.53
	Average	66.83	46.1	69.22	59.3	0.52
After preprocessing	Benign	99.6	99.8	99.5	97.9	0.99
	Malignant	99.8	99.8	99.6	98.3	1.0
	Normal	99.4	99.6	99.4	98.1	0.99
	Average	99.6	99.7	99.5	98.1	0.99

The bold values is the key takeaways or summarized results from the data

Table 3 The INbreast dataset results were applied using the ensemble pre-trained CNN and HGS optimizer

CNN	Class	Performance of the classifier				
		Accuracy (%)	Sensitivity	Specificity	Precision	AUC
Before preprocessing	Benign	66.6	66.8	70.1	56.8	0.55
	Malignant	66.8	63.9	67.8	60.3	0.52
	Normal	68.5	68.2	69.8	61.1	0.53
	Average	67.3	66.3	69.2	59.4	0.53
After preprocessing	Benign	99.8	100	99.8	98.3	1.0
	Malignant	100	100	100	98.1	1.0
	Normal	99.8	99.9	99.7	97.9	1.0
	Average	99.8	99.9	99.8	98.1	1.0

The bold values is the key takeaways or summarized results from the data

The results of the proposed ensemble model when GWO optimizes the model before and after preprocessing are

presented in Table 2. From this table, it can be observed that before preprocessing: (1) the suggested model

achieves the best accuracy, sensitivity, and precision when classifying normal cases with 68.1%, 49.8%, and 60.9%, respectively. (2) The best specificity is achieved when classifying malignant cases with 69.9%. On the other hand, after preprocessing, the accuracy, specificity, precision, and AUC reached 99.8%, 99.6%, 98.3%, and 1.0, respectively, in the malignant class.

When the ensemble model is optimized with the HGS algorithm, the model achieves the best results in almost all values, as illustrated in Table 3. The model achieved the best results for classifying normal cases before preprocessing with 68.5%, 68.2%, 69.8%, 61.1%, and 0.53 for accuracy, sensitivity, specificity, precision, and AUC, respectively. After preprocessing, the best results achieved in classifying normal cases were 99.8%, 99.9%, 99.7%, 97.9%, and 1.0 on the same performance metrics.

When compared to the malignant data, the ensemble network performs the best, with an overall performance of 100% for accuracy, sensitivity, specificity, and AUC when the hyperparameters are adjusted using the HGS algorithm. On the other hand, when the hyperparameters are adjusted using the GWO algorithm, the classification performance achieves 99.8%, 99.8%, 99.6%, 98.3%, and 1.0 for accuracy, sensitivity, specificity, precision, and AUC, respectively. When compared with the normal data, it achieved 99.8%, 99.9%, 99.7%, 97.9%, and 1.0 for accuracy, sensitivity, specificity, precision, and AUC, respectively, when optimized with HGS while achieving 99.4%, 99.6%, 99.4%, 98.1%, and 0.99 with GWO, with the same metrics.

The training and loss curves for the optimized ensemble model with GWO are shown in Figs. 8 and 9. The training and loss curves for the optimized ensemble model with HGS are shown in Figs. 10 and 11. The results indicate that the proposed model achieves the best average results with the HGS optimizer with 99.8%, 99.9%, 99.8%, 98.1%, and 1.0 for accuracy, sensitivity, specificity, precision, and



Fig. 8 The training curve for the optimized ensemble model with GWO

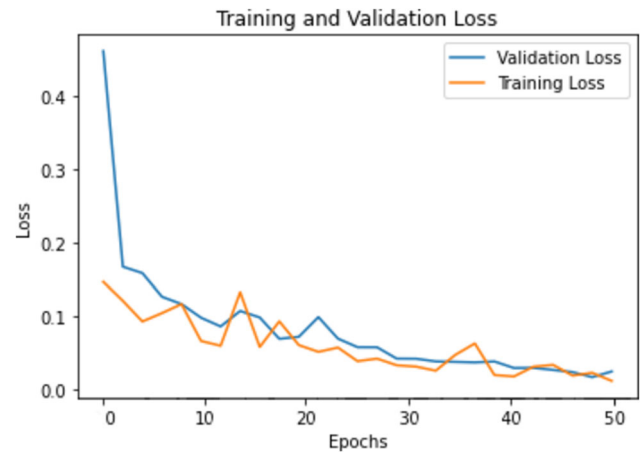


Fig. 9 The loss curve for the optimized ensemble model with GWO



Fig. 10 The training curve for the optimized ensemble model with HGS

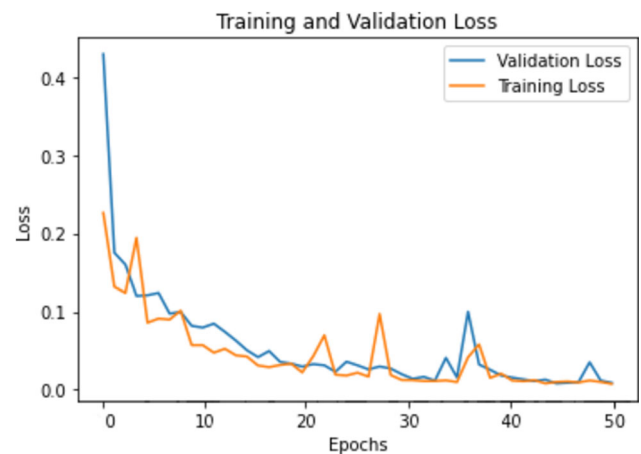


Fig. 11 The loss curve for the optimized ensemble model with HGS

AUC. On the same dataset, the experiments are compared with the other five of the most recently related models in Table 4. The analysis of the results proves that the

Table 4 Comparison between related works and the presented model based on the INbreast dataset

Method	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	AUC
Akselrod-Ballin et al. [13]	78	–	–	–	–
Al-Antari et al. [14]	95.3	–	–	–	–
Chakravarthy et al. [15]	96.6	–	–	–	–
Abeer et al. [16]	97.1	96.3	97.9	–	0.988
Abeer et al. [17]	99.2	98.8	99.1	96	0.998
The proposed	99.8	99.9	99.8	98.1	1.0

The bold values is the key takeaways or summarized results from the data

presented model achieved the best results in terms of accuracy, sensitivity, specificity, precision, and AUC.

5 Conclusion

In this paper, a novel optimized ensemble DL model-based meta-heuristic algorithm is proposed for enhancing the classification results on the INbreast mammographic dataset to help doctors in tumor detection and diagnosis more accurately. The INbreast images were classified into benign, malignant, and normal classes. The original dataset was pre-processed to improve tumor contract in mammographic images and determine the cancerous regions more precisely. Data augmentation was also applied to increase the dataset size to improve the classification result. The optimized ensemble model achieved the best accuracy, sensitivity, precision, F-score, and AUC, and contrasted with alternative models. Finally, it can be said that when compared to other current methods, a noticeable improvement may be obtained in the optimized ensemble CNN employing meta-heuristic algorithms in the screening process. The results demonstrated 99.8% accuracy, 99.9% sensitivity, 99.8 specificity, 98.1% precision, and 1.0 AUC. Comparing this model to the other related methods, it is better.

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